# Compoenent State

# Agenda

1. What is state
2. Defining state
3. Using state
4. Modifying state
5. Comparison of state and props

# 1. What is state

- State is the place where the data is maintained required by a component.
- Class components has a built-in state object.
- The state object can contain as many properties as required.
- Class components can have state but functional components can not have state but from version 16 onwards, by using hook functionality, state can even be mainatained in functional components.

# 2. Defining state

- Defining state in class component
  - Component's initial state can be added in a class constructor which assigns an initial values using this.state.
- Defining state in functional component.
  - To define a state in the functional components, useState hook can be used.
  - The hook accepts initial value and returns an array with two values: the current state and a function to update (setter function) it.
  - It can be used like this:
    ```
    const [state, setState] = useState(initialValue);
    ```

# 3. Using state

- Using state in class component
    - State object can be accessed anywhere in the class component by using the this.state.propertyname syntax
- Using state in functional component.
    - State can be used as a simple variable which can be rendered in JSX using curly braces syntax i.e. in the form of javascript expression.

# 4. Modifying state

- Modifying state in class component
  - To change a value in the state object, use this.setState() method.
- Modifying state in functional compoenent
  - State can be modified by calling it's setter method created in useState declaration.
- When the state is modified, component will get re-rendered displaying the latest state.

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# 4. Comparison between state and props

| Props | State |
|---|---|
| Props are read only and immutable | State is mutable. |
| Props are external and controlled by whatever renders the component. | The State is internal and controlled by the React Component itself. |
| Props can be accessed by the child component. | States can be used for rendering dynamic changes with the component. |